# Lab 2: Greedy regret heuristics

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**Source code:**
https://github.com/aleksanderhanski/evolutionary-computation/tree/main/lab2-greedy-regret-heuristics

## 1 Description of the problem

We are given a set of nodes. Each node has $x$ and $y$ coordinates in a plane and a cost. The goal is to select exactly 50% of the nodes (rounded up if the number of nodes is odd) and form a Hamiltonian cycle through the selected nodes, so that the sum of the total length of the cycle and the total cost of the selected nodes is minimized.

Distances between nodes are Euclidean distances rounded mathematically to integers. The distance matrix is computed once, right after reading an instance. The optimization methods only use the distance matrix and the vector of node costs, never the coordinates.

For a solution given as an ordered list of nodes $(v_0, \dots, v_{k-1})$:

$$f(v_0, \dots, v_{k-1}) = \sum_{i=0}^{k-1} D[v_i][v_{(i+1) \bmod k}] + \sum_{i=0}^{k-1} cost[v_i]$$

Both instances, TSPA and TSPB, have 200 nodes, so every solution contains $k = 100$ nodes.

## 2 Pseudocode of the implemented algorithms

Notation: $D$ is the distance matrix, $cost$ the node costs, $n$ the number of nodes, $k = \lceil n/2 \rceil$. As in Lab 1, "nearest" is never the plain distance: the change of the objective function after an insertion always includes the cost of the inserted node.

### Insertion cost

Both new methods are built on the two insertion heuristics from Lab 1. They differ only in the set of places where a node $v$ can be inserted and in the cost $\Delta(v, p)$ of inserting it at place $p$.

**Nearest neighbor (any position)** – the path is open while it is being built, the edge closing the cycle only appears in the final solution. The places are $p = 0, \dots, |P|$:

$$\Delta(v, p) = \begin{cases} D[v][P_0] + cost[v] & p = 0 \text{ (before the first node)} \\ D[P_{|P|-1}][v] + cost[v] & p = |P| \text{ (after the last node)} \\ D[a][v] + D[v][b] - D[a][b] + cost[v] & \text{otherwise, } a = P_{p-1},\ b = P_p \end{cases}$$

**Greedy cycle** – the solution is a closed cycle all the time, so the places are the $|P|$ edges $(a, b)$ of the cycle, including the edge (last, first):

$$\Delta(v, (a, b)) = D[a][v] + D[v][b] - D[a][b] + cost[v]$$

### Regret and the weighted criterion

For every node $v$ not in the solution let $\Delta_1(v)$ be its smallest and $\Delta_2(v)$ its second smallest insertion cost over all places. The 2-regret is

$$\mathit{regret}(v) = \Delta_2(v) - \Delta_1(v),$$

i.e. how much we lose if we do not insert $v$ now and its best place gets taken. Both methods select a node by a score, and the selected node is always inserted at its best place:

- **Greedy 2-regret**: $\mathit{score}(v) = \mathit{regret}(v)$,
- **Weighted sum**: $\mathit{score}(v) = w_r \cdot \mathit{regret}(v) - w_b \cdot \Delta_1(v)$, with equal weights $w_r = w_b = 1$ by default.

The weighted criterion with $w_r = 0$ is the plain greedy method from Lab 1, and with $w_b = 0$ it is the pure 2-regret heuristic. Ties in the score are broken by the smaller $\Delta_1$, then by the lower node index.

**Algorithm 1** Greedy regret heuristic (used for all four new methods)

**function** RegretHeuristic($D$, *cost*, *start*, $w_r$, $w_b$)
  $P \leftarrow [\mathit{start}]$
  **while** $|P| < k$ **do**
    **for all** nodes $v \notin P$ **do**
      **for all** places $p$ of $P$ **do** ▷ path or cycle places, see above
        compute $\Delta(v, p)$
      **end for**
      $\Delta_1(v), p^*(v) \leftarrow$ smallest $\Delta(v, \cdot)$ and its place
      $\Delta_2(v) \leftarrow$ second smallest $\Delta(v, \cdot)$ ▷ $= \Delta_1(v)$ if only one place
      $\mathit{score}(v) \leftarrow w_r \cdot (\Delta_2(v) - \Delta_1(v)) - w_b \cdot \Delta_1(v)$
    **end for**
    $v^* \leftarrow \arg\max_v \mathit{score}(v)$
    insert $v^*$ into $P$ at place $p^*(v^*)$
  **end while**
  **return** $P$
**end function**

The four methods are:

- NN (any position) 2-regret: path places, $w_r = 1, w_b = 0$,
- NN (any position) weighted: path places, $w_r = 1, w_b = 1$,
- Greedy cycle 2-regret: cycle places, $w_r = 1, w_b = 0$,
- Greedy cycle weighted: cycle places, $w_r = 1, w_b = 1$.

In the first step every place gives the same insertion cost: the path [*start*] has two places (before and after) with equal cost, and the one-node cycle has a single degenerate edge (*start*, *start*). All regrets are therefore 0 and the tie-break picks the node with the smallest $\Delta_1$, as in Lab 1.

## 3 Results of the computational experiment

For each greedy method 200 solutions were generated, one starting from each node. 200 random solutions were generated with a fixed seed (42). The results of the Lab 1 methods were recomputed in the same run and are identical to the previous report. Values are min / average / max of the objective function, the best value in each column is in bold.

| Method | TSPA min | TSPA avg | TSPA max |
|---|---|---|---|
| Random | 240892 | 262956.1 | 289850 |
| NN (end only) | 84362 | 87476.6 | 90075 |
| NN (any position) | 72328 | 74306.5 | 77022 |
| Greedy cycle | 72590 | 73743.7 | **75281** |
| NN (any position) 2-regret | 110950 | 121360.3 | 135771 |
| NN (any position) weighted | 72419 | 75549.4 | 78504 |
| Greedy cycle 2-regret | 112806 | 120378.8 | 128089 |
| Greedy cycle weighted | **72011** | **73353.5** | 75532 |

| Method | TSPB min | TSPB avg | TSPB max |
|---|---|---|---|
| Random | 186840 | 206333.8 | 227175 |
| NN (end only) | 56007 | 59397.3 | 63166 |
| NN (any position) | 52915 | 59476.2 | 66164 |
| Greedy cycle | 55655 | 59650.8 | 63734 |
| NN (any position) 2-regret | 68761 | 73649.8 | 79756 |
| NN (any position) weighted | **50485** | **55698.9** | **61406** |
| Greedy cycle 2-regret | 68150 | 72484.8 | 77274 |
| Greedy cycle weighted | 52731 | 57863.3 | 63370 |

### Other weights

We also ran the weighted criterion with other weights $(w_r, w_b)$, 200 solutions each. $(1, 0)$ is the pure 2-regret and $(0, 1)$ the plain greedy method; the latter differs from the Lab 1 numbers by less than 0.1% because ties are broken in a different order. Average objective, best value per row in bold:

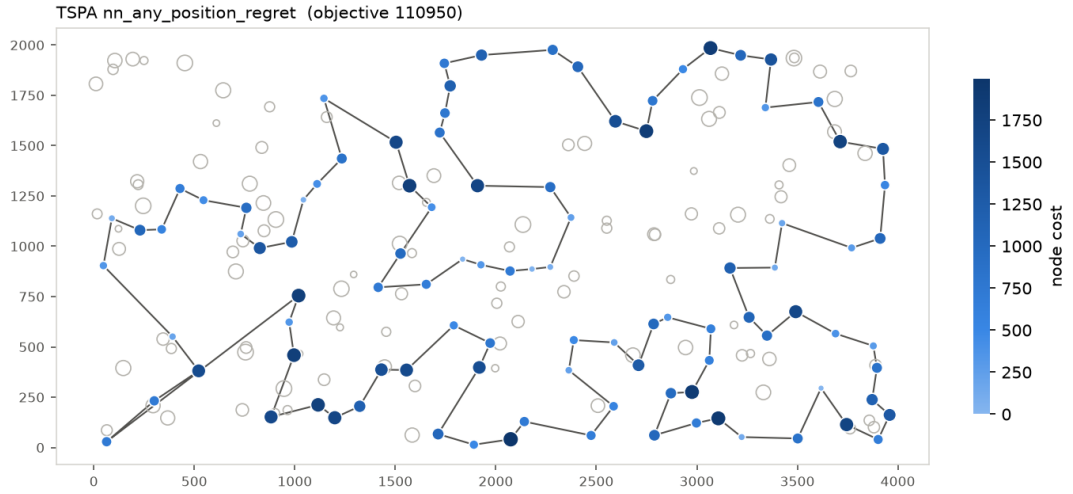
| Method | $(1, 0)$ | $(2, 1)$ | $(1, 1)$ | $(1, 2)$ | $(0, 1)$ |
|---|---|---|---|---|---|
| TSPA NN (any position) | 121360.3 | 76408.1 | 75549.4 | 75279.7 | **74347.0** |
| TSPA Greedy cycle | 120378.8 | 74277.2 | 73353.5 | **73189.0** | 73793.4 |
| TSPB NN (any position) | 73649.8 | **54710.1** | 55698.9 | 56150.8 | 59475.5 |
| TSPB Greedy cycle | 72484.8 | 59659.1 | 57863.3 | **55714.3** | 59649.9 |

## 4 Visualization of the best solutions

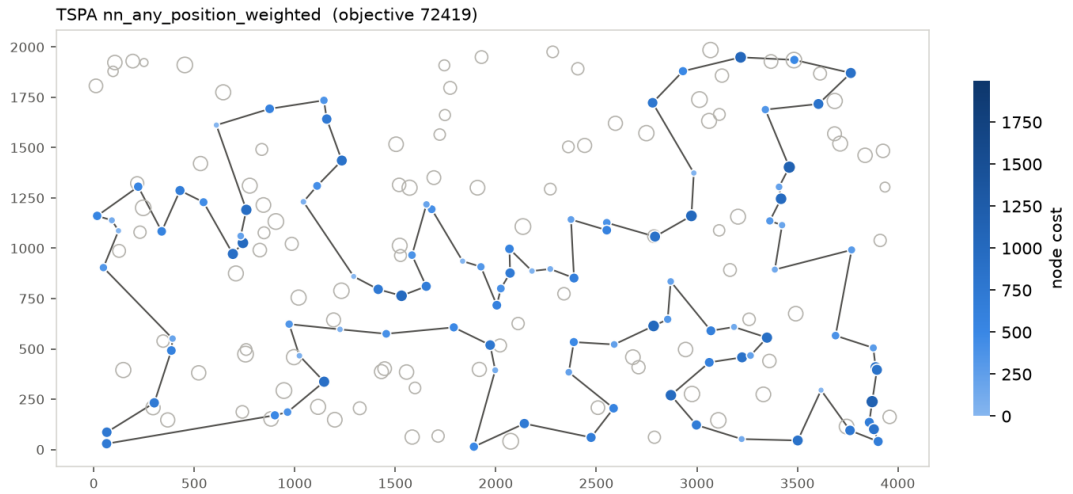
Each panel shows the best solution of one method. Selected nodes are filled, and both their size and color show the node cost (bigger and darker means more expensive). Nodes that are not selected are drawn as hollow grey circles, with size still showing their cost. Plots of the Lab 1 methods are in the previous report.

# TSPA

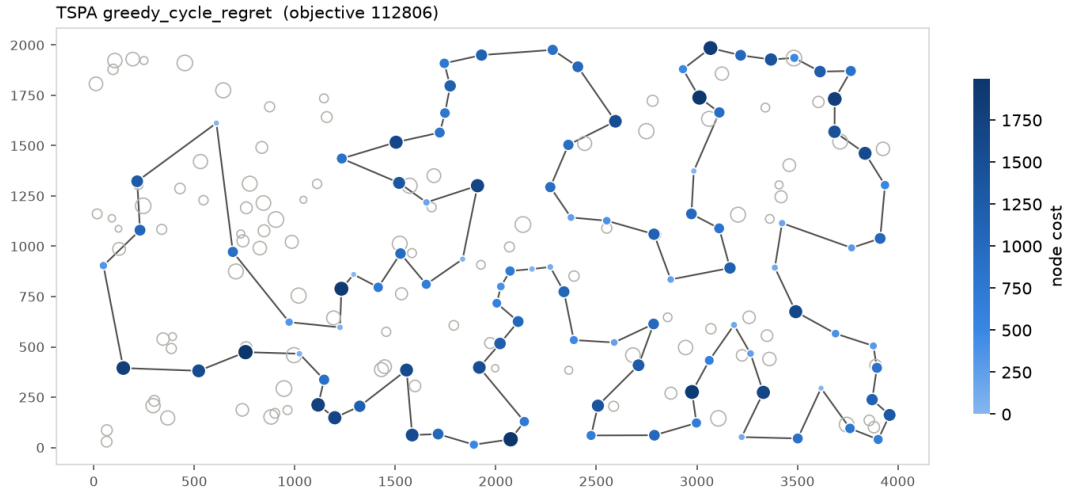
hollow = not selected, bigger/darker = higher cost



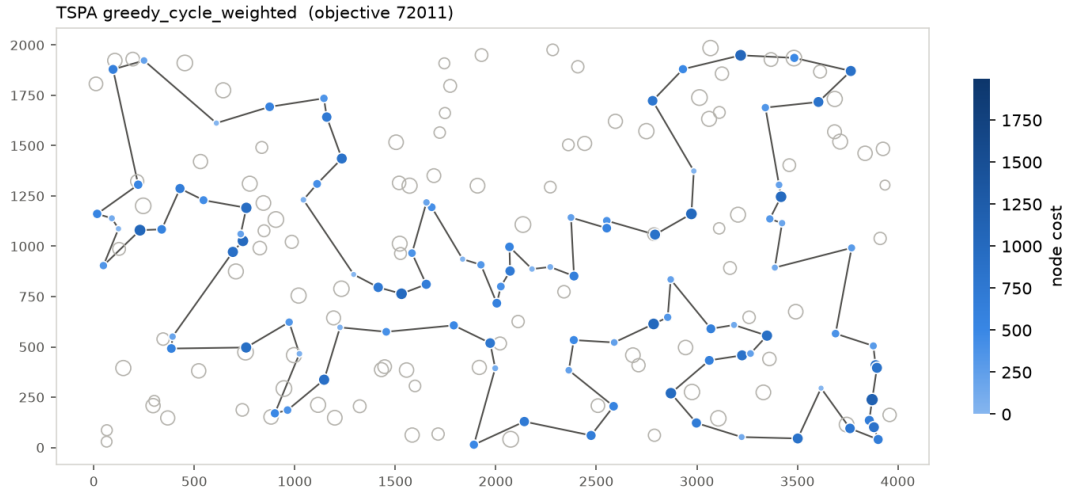

hollow = not selected, bigger/darker = higher cost



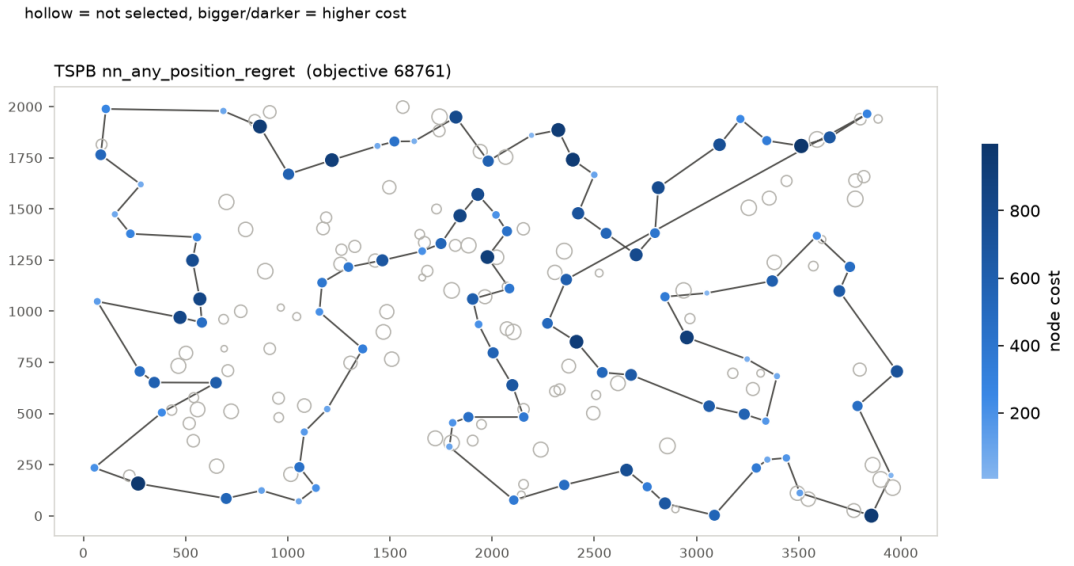
hollow = not selected, bigger/darker = higher cost

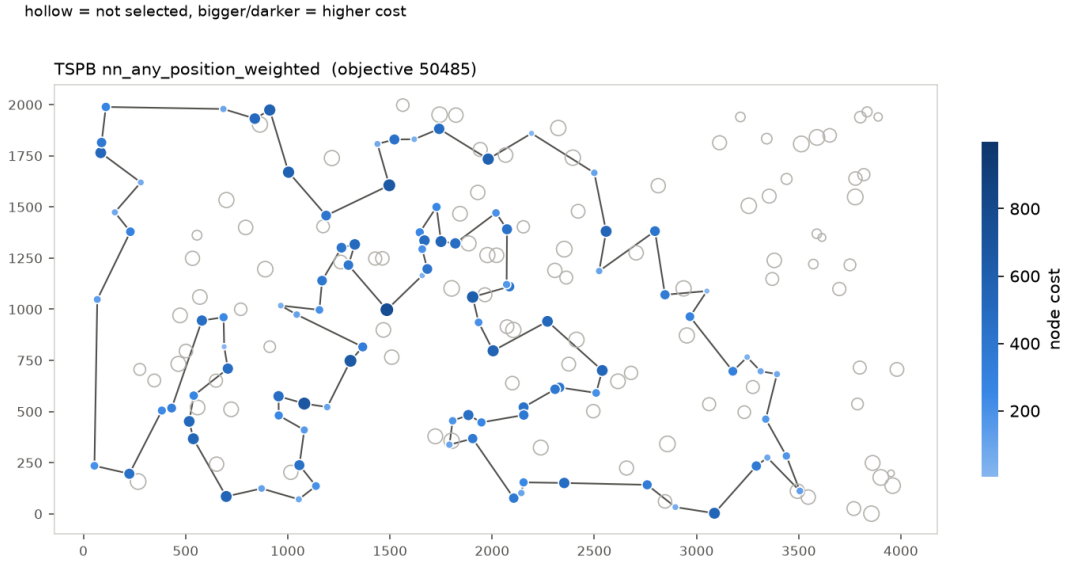


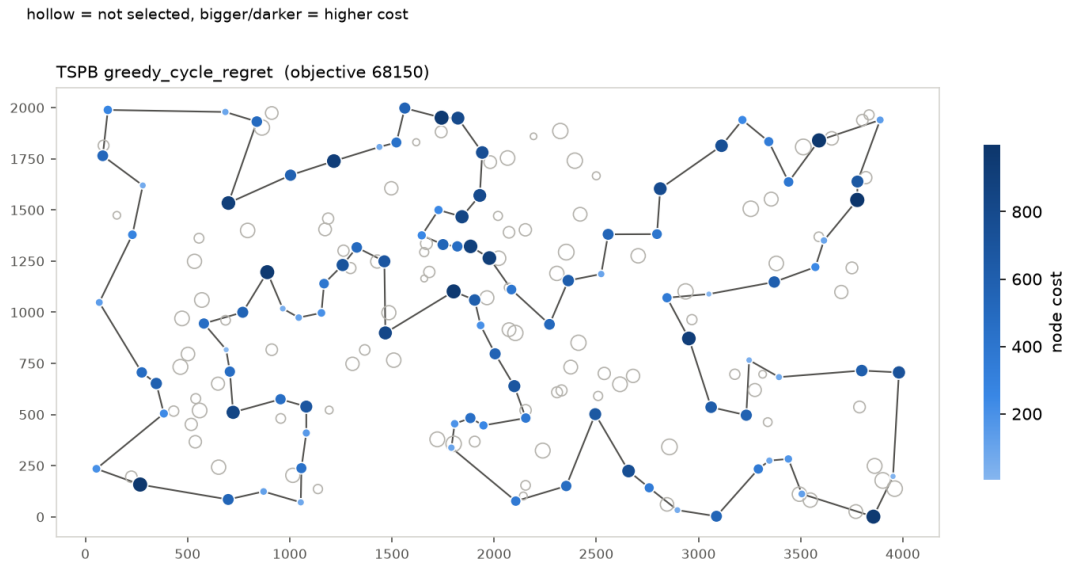

hollow = not selected, bigger/darker = higher cost

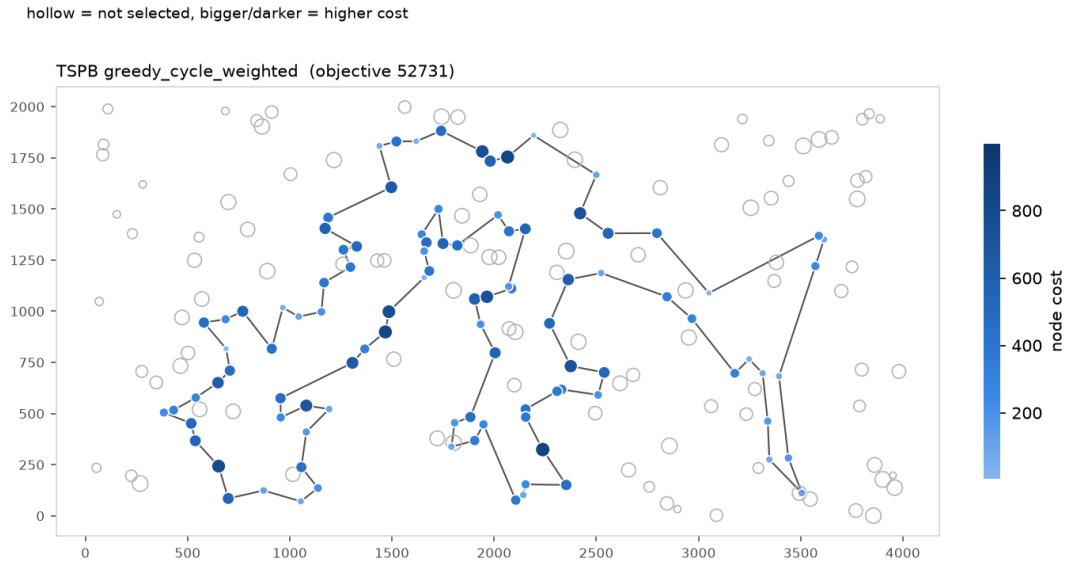


# TSPB


hollow = not selected, bigger/darker = higher cost
TSPB nn_any_position_regret (objective 68761)
node cost
200
400
600
800
0
250
500
750
1000
1250
1500
1750
2000
0
500
1000
1500
2000
2500
3000
3500
4000


hollow = not selected, bigger/darker = higher cost
TSPB nn_any_position_weighted (objective 50485)
node cost
200
400
600
800
0
250
500
750
1000
1250
1500
1750
2000
0
500
1000
1500
2000
2500
3000
3500
4000




# 5 Best solutions

Lists of node indices, starting from 0, in the order they are visited. The cycle closes from the last node back to the first. Best solutions of the Lab 1 methods are unchanged and listed in the previous report.

## TSPA

### NN (any position) 2-regret (objective 110950)

104, 69, 120, 97, 168, 149, 11, 49, 108, 152, 100, 123, 148, 162, 132, 76, 4, 157, 18, 173, 16, 156, 28, 185, 67, 144, 179, 177, 36, 33, 143, 190, 139, 112, 5, 115, 175, 39, 154, 34, 109, 186, 136, 197, 163, 19, 178, 101, 155, 21, 147, 37, 47, 50, 93, 81, 43, 113, 117, 130, 159, 170, 75, 26, 78, 122, 107, 64, 57, 105, 182, 111, 138, 165, 151, 150, 176, 72, 65, 134, 84, 58, 158, 192, 181, 118, 114, 25, 121, 73, 45, 70, 96, 10, 183, 166, 32, 40, 30, 23

### NN (any position) weighted (objective 72419)

171, 2, 189, 187, 30, 141, 87, 45, 73, 135, 114, 181, 192, 158, 58, 84, 134, 72, 176,

174, 150, 106, 113, 89, 42, 151, 138, 182, 57, 64, 107, 88, 78, 193, 153, 75, 170, 146, 159, 130, 47, 93, 50, 3, 169, 1, 59, 178, 101, 55, 29, 163, 136, 186, 164, 142, 8, 9, 54, 33, 131, 36, 177, 52, 179, 53, 140, 144, 67, 16, 51, 35, 185, 102, 28, 180, 132, 76, 4, 71, 157, 91, 80, 100, 123, 83, 160, 152, 108, 49, 60, 116, 149, 63, 168, 97, 127, 69, 129, 104

**Greedy cycle 2-regret** (objective 112806)

166, 183, 10, 70, 22, 25, 114, 118, 181, 121, 61, 41, 140, 53, 179, 177, 36, 99, 84, 134, 72, 65, 20, 192, 111, 182, 165, 151, 106, 89, 124, 57, 64, 107, 88, 78, 26, 75, 170, 159, 130, 117, 93, 50, 47, 37, 147, 6, 46, 86, 55, 17, 29, 19, 163, 197, 136, 195, 184, 164, 142, 85, 81, 174, 194, 9, 33, 143, 62, 34, 154, 39, 175, 115, 5, 112, 139, 18, 4, 196, 51, 190, 67, 185, 156, 28, 180, 191, 141, 30, 160, 80, 82, 11, 168, 188, 120, 125, 187, 189

**Greedy cycle weighted** (objective 72011)

60, 116, 149, 63, 168, 11, 49, 108, 152, 100, 123, 83, 160, 97, 127, 161, 30, 187, 171, 2, 189, 141, 87, 45, 73, 135, 114, 181, 192, 158, 58, 84, 134, 72, 176, 174, 150, 106, 113, 89, 42, 151, 138, 182, 57, 64, 107, 88, 78, 193, 153, 75, 170, 146, 159, 130, 47, 93, 50, 3, 169, 1, 178, 101, 55, 29, 163, 136, 186, 164, 142, 8, 9, 54, 33, 131, 36, 177, 52, 179, 53, 140, 144, 67, 16, 51, 35, 185, 102, 28, 180, 132, 76, 4, 71, 157, 91, 80, 48, 31

## TSPB

**NN (any position) 2-regret** (objective 68761)

94, 169, 199, 70, 190, 194, 77, 104, 144, 185, 79, 57, 83, 44, 89, 156, 164, 96, 198, 121, 195, 50, 7, 116, 37, 13, 149, 28, 145, 158, 138, 55, 177, 53, 65, 178, 103, 180, 132, 1, 108, 62, 175, 134, 45, 157, 60, 40, 64, 47, 38, 21, 168, 172, 8, 92, 166, 72, 46, 109, 3, 122, 151, 98, 25, 181, 73, 59, 85, 82, 88, 163, 99, 39, 17, 74, 20, 14, 2, 23, 110, 118, 131, 143, 148, 196, 78, 54, 162, 106, 127, 90, 153, 182, 150, 84, 66, 35, 27, 135

**NN (any position) weighted** (objective 50485)

65, 145, 28, 149, 13, 146, 37, 116, 147, 42, 50, 130, 63, 121, 198, 96, 48, 156, 89, 57, 185, 76, 104, 162, 54, 192, 80, 127, 129, 90, 153, 14, 2, 20, 74, 17, 171, 99, 88, 179, 87, 82, 51, 85, 59, 73, 167, 181, 152, 34, 24, 67, 66, 27, 98, 151, 122, 3, 184, 46, 72, 117, 8, 120, 133, 136, 172, 160, 69, 124, 21, 183, 30, 38, 47, 154, 140, 64, 33, 40, 141, 155, 71, 60, 157, 45, 134, 175, 62, 36, 197, 6, 115, 170, 125, 177, 29, 132, 174, 1

**Greedy cycle 2-regret** (objective 68150)

167, 181, 25, 98, 151, 122, 128, 4, 168, 183, 176, 38, 47, 140, 154, 161, 26, 177, 170, 115, 113, 155, 141, 60, 157, 134, 175, 62, 108, 1, 132, 103, 178, 65, 145, 149, 13, 37, 116, 147, 123, 50, 195, 121, 198, 58, 16, 164, 43, 166, 92, 120, 133, 8, 117, 95, 109, 3, 27, 135, 76, 185, 104, 77, 194, 190, 70, 187, 165, 105, 139, 193, 114, 10, 78, 54, 162, 106, 150, 182, 127, 90, 56, 131, 110, 23, 2, 14, 20, 74, 17, 171, 99, 163, 31, 88, 82, 85, 59, 73

**Greedy cycle weighted** (objective 52731)

36, 197, 132, 29, 6, 180, 115, 170, 177, 125, 26, 97, 154, 140, 47, 38, 21, 30, 183, 191, 130, 63, 121, 198, 96, 48, 43, 156, 32, 89, 57, 79, 185, 104, 54, 196, 114, 10, 90, 14, 2, 20, 153, 129, 127, 80, 192, 162, 76, 135, 27, 11, 66, 67, 24, 34, 152, 181, 126, 88, 179, 87, 82, 167, 51, 85, 59, 73, 98, 151, 122, 173, 3, 184, 86, 46, 72, 117, 8, 120, 133, 136, 172, 160, 69, 124, 4, 64, 33, 155, 71, 141, 40, 60, 157, 45, 134, 175, 62, 91

## 6 Solution checker

All sixteen best solutions (eight per instance, including the Lab 1 methods) were checked with the provided solution checker (`Solution checker.xlsx`), and every objective value matches. Our code additionally checks every generated solution for the number of nodes and for repeated nodes.

| Instance | Method | Our objective | Checker objective | Match |
|---|---|---|---|---|
| TSPA | Random | 240892 | 240892 | yes |
| TSPA | NN (end only) | 84362 | 84362 | yes |
| TSPA | NN (any position) | 72328 | 72328 | yes |
| TSPA | Greedy cycle | 72590 | 72590 | yes |
| TSPA | NN (any position) 2-regret | 110950 | 110950 | yes |
| TSPA | NN (any position) weighted | 72419 | 72419 | yes |
| TSPA | Greedy cycle 2-regret | 112806 | 112806 | yes |
| TSPA | Greedy cycle weighted | 72011 | 72011 | yes |
| TSPB | Random | 186840 | 186840 | yes |
| TSPB | NN (end only) | 56007 | 56007 | yes |
| TSPB | NN (any position) | 52915 | 52915 | yes |
| TSPB | Greedy cycle | 55655 | 55655 | yes |
| TSPB | NN (any position) 2-regret | 68761 | 68761 | yes |
| TSPB | NN (any position) weighted | 50485 | 50485 | yes |
| TSPB | Greedy cycle 2-regret | 68150 | 68150 | yes |
| TSPB | Greedy cycle weighted | 52731 | 52731 | yes |

## 7 Conclusions

- The pure 2-regret heuristic is clearly worse than the plain greedy methods: about 120 000 on average on TSPA (vs. 73 744 for greedy cycle) and about 73 000 on TSPB (vs. 59 651). The node cost is the same at every place, so it cancels out in $\Delta_2(v) - \Delta_1(v)$ and the regret does not see it at all. The tour lengths are similar (21 429 for the best greedy cycle 2-regret solution on TSPA vs. 21 570 for greedy cycle), but the selected nodes cost 91 377 instead of 51 020. The plots show this: the regret solutions contain many large, dark nodes.
- The weighted criterion (2-regret + best change, equal weights) fixes this and gives the best results of all methods so far. On TSPA greedy cycle weighted has the best minimum (72 011) and average (73 353.5, 0.5% below greedy cycle). On TSPB the gain is larger: NN (any position) weighted has the best minimum, average and maximum (50 485 / 55 698.9 / 61 406), its average is 6.4% below NN (any position), and greedy cycle weighted improves the greedy cycle average by 3.0%.
- The regret term mainly shortens the tour: the best greedy cycle weighted solution on TSPB has length 16 365 vs. 19 634 for greedy cycle, with almost the same node cost (36 366 vs. 36 021). Nodes whose good insertion places are about to disappear are inserted early, so the method leaves fewer long detours and crossings.
- The weights matter, and no single pair is the best everywhere. Putting more weight on the best change, $(1, 2)$, gave the best greedy cycle averages on both instances (73 189.0 and 55 714.3), while NN (any position) on TSPB was best with more weight on regret, $(2, 1)$. On TSPA, NN (any position) did not gain from the regret term on average. Equal weights are a reasonable default: never the worst choice, and always far better than the pure 2-regret.

- The regret methods cost almost nothing extra: they need the same insertion costs as the greedy methods, plus the second smallest value per node. Our vectorized implementation takes about 15 ms per solution.